

GROUP 6

GROW DATA ANALYST PROGRAM COHORT 2

How can public sector organizations use data analytics to drive smart city initiatives, such as traffic management and urban planning, and what are the key barriers to adoption?

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Why is Smart City Important?



Image Credit: Rethinking The Future

In this context, smart city is about using innovative technologies and methods to make the transport system more efficient with a well-planned settlement. By adopting this approach, we can increase productivity, save time and improve the overall quality of life.

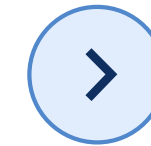
Improve quality of life

Save time

Increased productivity

Data Analytics in Transport Planning

This involves collecting, organizing, interpreting data related to transportation in order to validate informed decisions or hypothesis



Assessing Data Needs

Identifying needs and relevant gaps, eg. from traffic trends and traveler habits



Data collection

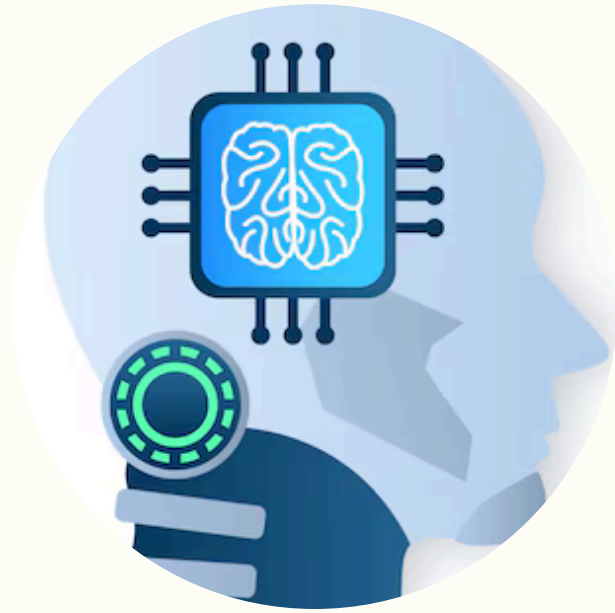
Gathering data from GPS tracking, traffic sensors, historical data, IoT sensors, bookings from clients and survey from travels



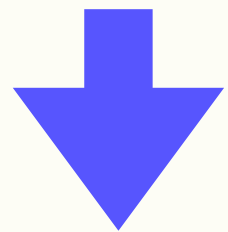
Use of predictive analysis

Insights from predictive analysis to manage operations and logistics seamlessly

Automation in Transport System



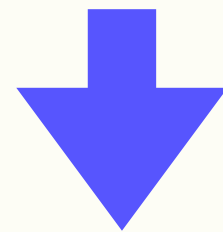
Machine Learning



Traffic prediction



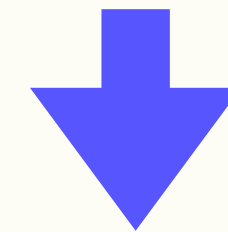
GIS Mapping



Route optimization



Data Analytics



Logistics planning

Urban Planning



Data on Community settlement

Obtaining Data on settlement from
Land Use and Spatial Planning
(LUSPA)



Data visualization

Use of scatter plots, heat maps,
bubble charts, etc.



Diagnostic Analysis

Accurate diagnosis of the spatial
quality can help solve issues of
overcrowding

Barriers to Adoption

Data quality

Data may be compromised and may be misleading

Data Security

Users unwilling to share information or data about themselves

Skilled workforce

Inadequate skilled labour force to carry out the analytical and other technological aspects

Sustainability Issues

Having to sustain the implemented initiatives due to lack of interest, shift in government priorities, etc...

Budgetary constraint

Not having enough funding to implement the initiative

Stable internet

Issues with sustainable energy - electricity, internet may make it adoption or its use to be unpleasant

Strategies

Citizen Sensitization

Creating awareness and engaging users on the adoption of the technology



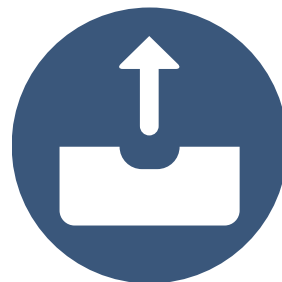
Uniform Database

Have a centralized database with strict supervision



Invest in Human Resources

Investment in human resources to solve issue of inadequate skilled labour



Data Privacy Policy

Implement data privacy policies to assure users of their confidentiality



References

- Forbytes (2024). The Power of Data Analytics in Transportation: Key Benefits and Use Cases Retrieved from [Link](#).
- JS Morlu (2024). Smart Cities in Ghana: The Role of Technology in Urban Development. Retrieved from [Link](#).
- UNCDF (2023). Urban Economic Recovery and Resilience. Diagnostic and Planning tool. Retrieved from [Link](#).

THANK

YOU!

